```
hozaifa@DESKTOP-64LGA3H:~$ ls
gencode_project
hozaifa@DESKTOP-64LGA3H:~$ cd gencode_project/
hozaifa@DESKTOP-64LGA3H:~/gencode_project$ ls
'1_control_18S_2019_minq7 - Copy.fastq?download=1'   Biotecha.ipynb   Sequences.py   Test.ipynb   bio_env   gencode.v50.pc_transcripts.fa   spike.gb
hozaifa@DESKTOP-64LGA3H:~/gencode_project$ source bio_env/bin/activate
(bio_env) hozaifa@DESKTOP-64LGA3H:~/gencode_project$ python
Python 3.14.4 (main, Aug 20 2026, 10:41:58) [GCC 15.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>>
>>>
>>> from Bio import SeqIO

>>>
>>> filename = 'gencode.v50.pc_transcripts.fa'
>>> seq = SeqIO.parse(filename, 'fasta')
>>>
>>> for record in seq :
...
...
...     break
...
>>> L =[]
>>>
>>> for record in seq :
...     L.append(record)
...
>>> L[3].name
'ENST00001114008.1|ENSG00000187634.15|OTTHUMG00000040719.11|-|SAMD11-219|SAMD11|2523|UTR5:1-537|CDS:538-1857|UTR3:1858-2523|'
>>> print(L[3].seq)
```

Ubuntu

```
(bio_env) hozaifa@DESKTOP-64LGA3H:~/gencode_project$ python
Python 3.14.4 (main, Aug 20 2026, 10:41:58) [GCC 15.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>>
>>>
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>>> print(L[3].seq)
AAATGGGCCAAGTTGTGGCGGCGGCGTCGGCGGCGGAGTCTCCCAAGTCCCCGCCGGGCGGGCGCGCGCCAGTGGACGCGGGTGCACGACTGACGCGGCCCGGGCGGCGGGGCGGGGGCTTGGGACCCCCGAGAGGGGCGGGGACTCCGCGACTCCTCGCTGCCGGGCTC
GGCCTGGCGGGTGGGTCGGCGAGCCGGGCGTGGGACTGCCCCGGGCGCGGGCGCTGGTGGCCGGGGCGCGGGACTCCAGACGCCCCGGGGAGCCCCGAGGCCCTGGAACTGCGGCGCTCGGCGAGTCGATCCGGGATCGATAGCAGCTCCATGTCTCCGGCCTCTGAGGC
CCCGCCGGCCGGCTGGGCAGTCCGGGGAGGCCTGGCGGGCGGCGCGTAGGCGGCGGCTGCGGGCGCCGGGGCGCACTAGCGGACGGCGTGGGCGCGCGGCCAGGCGCCTCCCCGGCCCCCGCGACCCAACTCCAGCCCGGGCCGGAATAAGTTGCTGCCGCCGGCGGAGA
GCGGGGCTGCGGAGCCACCGGGGCGCCATGCCGGCGGTCAAGAAGGAGTTCCCGGGCCGCGAGGACCTGGCCCTGGCTCTGGCCACGTTCCACCCGACCCTGGCCGCGCTGCCGCTGCCGCCGCTGCCAGGCTACCTGGCGCCACTGCCCGCGGCGGCCGCCCTCCCCCC
GGCCGCCTCGCTGCCCGCCTCGGCCGCCGGTTACGAGGCTCTGCTGGCCCCGCCGCTCCGCCCCCCGCGCGCCTACCTCAGCCTGCACGAGGCCGCCCCGCACCTCCACCTGCCCAGGGACCCGCTGGCCCTCGAGCGCTTCTCGGCCACCGCGGCCGCGGCCCCGGATT
TCCAGCCGCTGCTGGACAACGGCGAGCCGTGCATCGAGGTGGAGTGCGGCGCCAACCGCGCGCTGCTCTACGTGCGCAAACTCTGCCAGGGCAGCAAGGGCCCGTCCATCCGCCACCGCGGCGAGTGGCTCACGCCCAACGAGTTCCAGTTCGTCAGCGGCCGCGAGACG
GCCAAGGACTGGAAGCGCAGCATCCGCCACAAAGGGAAAAGTCTGAAGACGCTTATGTCCAAGGGGATCCTGCAGGTGCATCCTCCGATCTGCGACTGCCCGGGCTGCCGAATATCCTCCCCGGTGAACCGGGGGCGGCTGGCAGACAAGAGGACAGTCGCCCTGCCTGC
CGCCCGGAACCTGAAGAAGGAGCGAACTCCCAGCTTCTCTGCCAGCGATGGTGACAGCGACGGGAGTGGCCCCACCTGTGGGCGGCGGCCAGGCTTGAAGCAGGAGGATGGTCCGCACATCCGTATCATGAAGAGAAGAGTCCACACCCACTGGGACGTGAACATCTCTT
TCCGAGAGGCGTCCTGCAGCCAGGACGGCAACCTTCCCACCCTCATATCCAGCGTCCACCGCAGCCGCCACCTCGTTATGCCCGAGCATCAGAGCCGCTGTGAATTCCAGAGAGGCAGCCTGGAGATTGGCCTGCGACCCGCCGGTGACCTGTTGGGCAAGAGGCTGGGC
CGCTCCCCCCGTATCAGCAGCGACTGCTTTTCAGAGAAGAGGGCACGAAGCGAATCGCCTCAAGCAGAGGCGCTGCTGCTGCCGCGGGAGCTGGGGCCCAGCATGGCCCCGGAGGACCATTACCGCCGGCTTGTGTCAGCACTGAGCGAGGCCAGCACCTTTGAGGACCC
TCAGCGCCTCTACCACCTGGGCCTCCCCAGCCACGGTGAGGACCCACCCTGGCATGATCCCCCTCATCACCTCCCCAGCCACGGTGAGGACCCACCCTGGCATGATCTCCCCTCATCACCTCCCCAGCCACATGTACTCGGCCATTCCTGTTGCTGAGGCCCTGCTGACA
CCAAGGCCAGGCTGGATGCAGGTCCCTCTGCCACACGTCCTGCCCCATGCCCCCTGGGGCGGGCCACACCTCCATGTCCCCTAGGTCCCCAGGGTCATGACTAGCTCACATTTTATATAGAGAGAAATGGAGTCTGGGGTGGACCCAGGTGAGGGTGGGCAGTGGGCATG
TCAGCAGCACCCCCCGAGGAGAGCAAGCTCCTGGACCCTGTGGTCTGTGAGTCGTCTATGCAGCCAGTGGACGCCGACCTGCCAGACGCCTGCCCCAGGAGCCTGGGGAGGGGCAGTGAGCAGAAAGGCCGGGCTGGGTGCAGTGGGCACTTGGCCACCAGGACTCCCCA
GGTGCTGAAGAGACGCCAGCTGGAGGGGCTGCCCCTTCCCCCGGGTCGGCCCTGACCCTGTCCACCCCACCTCAGGACGTTCTCCAGGGGTCCCTCCGGGATGCACTCGGACCCCCTGCCCGCTGCACTCAGCCTCCCAGGCCCCAGCCGCCCGCCTGGCAGGGGAGCTT
GGCTTTTCGGGCTAGAGGTGGGTGGGGGCGCCGGGAAAGGAGGCAGGATTCCTCACACCAGGCACCGTCCCCCAGGGCAGCTCAGGCACCAAGAGCCTGAATAATTCACCAAATGTTAATAATGTAAAAATCCTCCTTTTTAA
>>>
```

Download BLAST
Get BLAST databases and executables

Use BLAST API
Call BLAST from your application

Use BLAST in the cloud
Start an instance at a cloud provider

Specialized searches

SmartBLAST
Find proteins highly similar to your query

Primer-BLAST
Design primers specific to your PCR template

Global Align
Compare two sequences across their entire span (Needleman-Wunsch)

CD-search
Find conserved domains in your sequence

IgBLAST
Search immunoglobulins and T cell receptor sequences

VecScreen
Search sequences for vector contamination

CDART
Find sequences with similar conserved domain architecture

Multiple Alignment
Align sequences using domain and protein constraints

Primer designing tool

https://www.ncbi.nlm.nih.gov/tools/primer-blast/index.cgi?LINK_LOC=BlastHome

Chat

An official website of the United States government Here's how you know

National Library of Medicine
National Center for Biotechnology Information

Log in

Primer-BLAST

A tool for finding specific primers

Finding primers specific to your PCR template (using Primer3 and BLAST).

Primers for target on one template | Primers common for a group of sequences

Retrieve recent results | Publication | Tips for finding specific primers | Save search parameters | Reset page

PCR Template

Enter accession, gi, or FASTA sequence (A refseq record is preferred) Clear

>ENST00001114008.1|ENSG00000187634.15|OTTHUMG00000040719.11|-|SAMD11-219|SAMD11|2523|UTR5:1-537|CDS:538-1857|UTR3:1858-2523|

Or, upload FASTA file Choose File No file chosen

Range Clear

	From	To
Forward primer		
Reverse primer		

Primer Parameters

Use my own forward primer (5'->3' on plus strand) Clear

Use my own reverse primer (5'->3' on minus strand) Clear

	Min	Max
PCR product size	70	1000
# of primers to return	10	

	Min	Opt	Max	Max T_m difference
Primer melting temperatures (T_m)	57.0	60.0	63.0	3

Primer Location Preference Prefer primers at 3' side of the template

Exon/intron selection

A refseq mRNA sequence as PCR template input is required for options in the section

Exon junction span No preference

CCACCCCACCTCAGGACGTTCTCCAGGGGTCCCTCCGGGATGCACTCGGACCCCCTGCCCGC
TGCACTCAGCCTCCCAGGCCCCAGCCGCCCGCCTGGCAGGGGAGCTTGGCTTTTCGGGCTAG
AGGTGGGTGGGGGCGCCGGGAAAGGAGGCAGGATTCCTCACACCAGGCACCGTCCCCCAGG
GCAGCTCAGGCACCAAGAGCCTGAATAATTCACCAAATGTTAATAATGTAAAAATCCTCCTTT
TTAA

Or, upload FASTA file: Choose File | No file chosen

	From	To
Forward primer	1	400
Reverse primer	2000	2406

Primer Parameters

Use my own forward primer (5'->3' on plus strand): Clear

Use my own reverse primer (5'->3' on minus strand): Clear

PCR product size: Min 70 | Max 3000

of primers to return: 10

Primer melting temperatures (T_m): Min 57.0 | Opt 60.0 | Max 63.0 | Max T_m difference 3

Primer Location Preference: Prefer primers at 3' side of the template

Exon/intron selection

A refseq mRNA sequence as PCR template input is required for options in the section

Exon junction span: No preference

Exon junction match: Min 5' match 7 | Min 3' match 4 | Max 3' match 8

Minimal and maximal number of bases that must anneal to exons at the 5' or 3' side of the junction

Intron inclusion: Primer pair must be separated by at least one intron on the corresponding genomic DNA

Intron length range: Min 1000 | Max 10000

Note: Parameter values that differ from the default are highlighted in yellow

Primer Pair Specificity Checking Parameters

Specificity check: Enable search for primer pairs specific to the intended PCR template

Search mode: Automatic

Database: Refseq mRNA

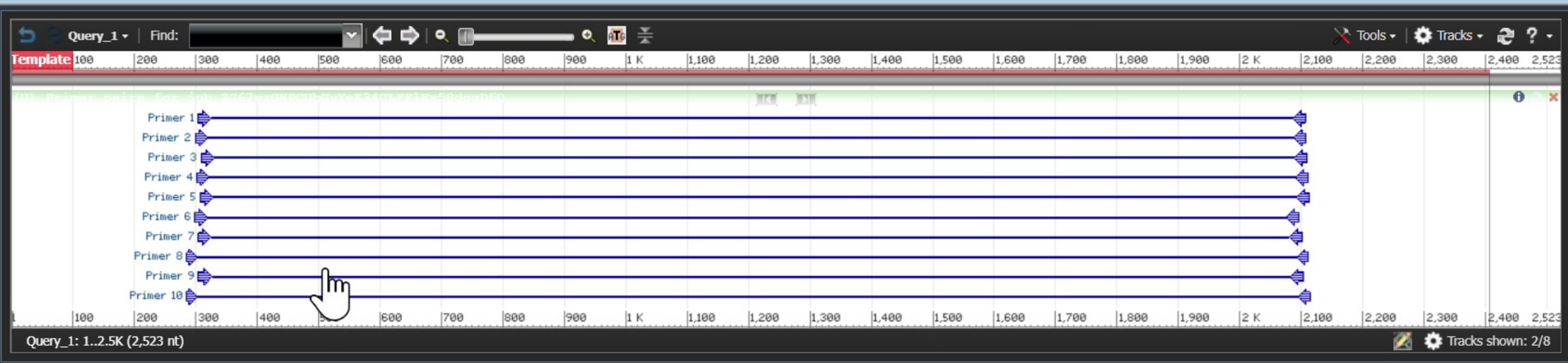
1857|UTR3:1858-2523|

Range 1 - 2406

Specificity of primers Primer pairs are specific to input template as no other targets were found in selected database: NCBI Messenger RNA Reference Sequences (Organism limited to Homo sapiens)

Other reports Search Summary

Graphical view of primer pairs



Detailed primer reports

Download primer pairs

Primer pair 1

	Sequence (5'->3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	GGATCGATAGCAGCTCCATGT	Plus	21	304	324	59.73	52.38	6.00	2.00
Reverse primer	TGGCTGCATAGACGACTCAC	Minus	20	2106	2087	59.83	55.00	4.00	0.00
Product length	1803								

Primer pair 2

— Detailed primer reports

Download primer pairs

Primer pair 1

	Sequence (5'->3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	GGATCGATAGCAGCTCCATGT	Plus	21	304	324	59.73	52.38	6.00	2.00
Reverse primer	TGGCTGCATAGACGACTCAC	Minus	20	2106	2087	59.83	55.00	4.00	0.00
Product length	1803								

Primer pair 2

	Sequence (5'->3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	TCCGGGATCGATAGCAGCTC	Plus	20	300	319	61.24	60.00	6.00	3.00
Reverse primer	CTGGCTGCATAGACGACTCA	Minus	20	2107	2088	59.54	55.00	4.00	2.00
Product length	1808								

Primer pair 3

	Sequence (5'->3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	GATAGCAGCTCCATGTCTCCG	Plus	21	309	329	60.34	57.14	5.00	2.00
Reverse primer	ACTGGCTGCATAGACGACTC	Minus	20	2108	2089	59.54	55.00	4.00	3.00
Product length	1800								

Primer pair 4

	Sequence (5'->3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	CGGGATCGATAGCAGCTCCA	Plus	20	302	321	61.53	60.00	6.00	2.00
Reverse primer	CCACTGGCTGCATAGACGAC	Minus	20	2110	2091	60.81	60.00	5.00	2.00
Product length	1809								

Primer pair 5

	Sequence (5'->3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	TCGATAGCAGCTCCATGTCTC	Plus	21	307	327	59.39	52.38	5.00	0.00
Reverse primer	GTCCACTGGCTGCATAGACG	Minus	20	2112	2093	60.81	60.00	5.00	2.00
Product length	1806								

Primer pair 6

	Sequence (5'->3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	ATCCGGGATCGATAGCAGCTC	Plus	21	299	319	61.56	57.14	7.00	3.00
Reverse primer	ACGACTCACAGACCACAGGG	Minus	20	2095	2076	61.46	60.00	3.00	1.00
Product length	1797								

Format Conversion
-Combine FASTA
-EMBL to FASTA
-EMBL Feature Extractor
-EMBL Trans Extractor
-Filter DNA
-Filter Protein
-GenBank to FASTA
-GenBank Feature Extractor
-GenBank Trans Extractor
-One to Three
-Range Extractor DNA
-Range Extractor Protein
-Reverse Complement
-Split Codons
-Split FASTA
-Three to One
-Window Extractor DNA
-Window Extractor Protein
Sequence Analysis
-Codon Plot
-Codon Usage
-CpG Islands
-DNA Molecular Weight
-DNA Pattern Find
-DNA Stats
-Fuzzy Search DNA
-Fuzzy Search Protein
-Ident and Sim
-Multi Rev Trans
-Mutate for Digest
-ORF Finder
-Pairwise Align Codons
-Pairwise Align DNA
-Pairwise Align Protein
-PCR Primer Stats
-PCR Products
-Protein GRAVY
-Protein Isoelectric Point
-Protein Molecular Weight
-Protein Pattern Find
-Protein Stats
-Restriction Digest
-Restriction Summary
-Reverse Translate
-Translate
Sequence Figures
-Color Align Conservation
-Color Align Properties
-Group DNA
-Group Protein
-Primer Map
-Restriction Map
-Translation Map
Random Sequences
-Mutate DNA
-Mutate Protein
-Random Coding DNA
-Random DNA Sequence
-Random DNA Regions
-Random Protein Sequence
-Random Protein Regions
-Sample DNA
-Sample Protein

Sequence Manipulation Suite:

PCR Primer Stats

New version available: Sequence Manipulation Suite 3 (SMS3)

PCR Primer Stats accepts a list of PCR primer sequences and returns a report describing the properties of each primer, including melting temperature, percent GC content, and PCR suitability. Use PCR Primer Stats to evaluate potential PCR primers.

Paste the raw sequence or one or more FASTA sequences into the text area below. Input limit is 5,000,000 characters. The maximum accepted primer length is 50 bases.

```
>primer1
GGATCGATAGCAGCTCCATGT
>primer2
```

Submit Clear Reset

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*You can mirror this page or use it off-line.

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Mon Jun 15 14:58:15 2026
Valid XHTML 1.0; Valid CSS

PCR Primer Stats results

```
Global settings:
-The primers do not have a 5'-phosphate group.
-Combined concentration of K+ and Na+ in the reaction = 50 millimolar.
-Mg+2 concentration in the reaction = 1.5 millimolar.
-Primer concentration in the reaction = 200 nanomolar.

------------------------------------------------------------
------------------------------------------------------------
General properties:
-------------------
                  Primer name: primer1
              Primer sequence: GGATCGATAGCAGCTCCATGT
              Sequence length: 21
                  Base counts: G=6; A=5; T=5; C=5; Other=0;
               GC content (%): 52.38
    Molecular weight (Daltons): 6446.25
                   nmol/A260: 4.89
             micrograms/A260: 31.55
          Basic Tm (degrees C): 54
   Salt adjusted Tm (degrees C): 49
Nearest neighbor Tm (degrees C): 64.93

PCR suitability tests (Pass / Warning):
-----------------------------------
             Single base runs: Pass
        Dinucleotide base runs: Pass
                       Length: Pass
                   Percent GC: Pass
         Tm (Nearest neighbor): Warning: Tm is greater than 58;
                     GC clamp: Pass
               Self-annealing: Warning: There are more than 3 self-annealing bases in a row;
                             :                GGATCGATAGCAGCTCCATGT
                             :                 ||||
                             : TGTACCTCGACGATAGCTAGG
           Hairpin formation: Pass
------------------------------------------------------------

------------------------------------------------------------
General properties:
-------------------
                  Primer name: primer2
              Primer sequence: TGGCTGCATAGACGACTCAC
              Sequence length: 20
                  Base counts: G=5; A=5; T=4; C=6; Other=0;
               GC content (%): 55.00
```

Sequence Manipulation Suite - Personal - Microsoft Edge

about:blank

```
                    nmol/A260: 4.89
              micrograms/A260: 31.55
          Basic Tm (degrees C): 54
   Salt adjusted Tm (degrees C): 49
Nearest neighbor Tm (degrees C): 64.93

PCR suitability tests (Pass / Warning):
-----------------------------------
              Single base runs: Pass
        Dinucleotide base runs: Pass
                       Length: Pass
                   Percent GC: Pass
         Tm (Nearest neighbor): Warning: Tm is greater than 58;
                     GC clamp: Pass
                Self-annealing: Warning: There are more than 3 self-annealing bases in a row;
                              :                GGATCGATAGCAGCTCCATGT
                              :                ||||
                              : TGTACCTCGACGATAGCTAGG
            Hairpin formation: Pass
-------------------------------------------------------------

-------------------------------------------------------------
General properties:
-------------------
                  Primer name: primer2
               Primer sequence: TGGCTGCATAGACGACTCAC
               Sequence length: 20
                  Base counts: G=5; A=5; T=4; C=6; Other=0;
               GC content (%): 55.00
    Molecular weight (Daltons): 6102.02
                    nmol/A260: 5.28
              micrograms/A260: 32.22
          Basic Tm (degrees C): 54
   Salt adjusted Tm (degrees C): 49
Nearest neighbor Tm (degrees C): 65.09

PCR suitability tests (Pass / Warning):
-----------------------------------
              Single base runs: Pass
        Dinucleotide base runs: Pass
                       Length: Pass
                   Percent GC: Pass
         Tm (Nearest neighbor): Warning: Tm is greater than 58;
                     GC clamp: Pass
                Self-annealing: Pass
            Hairpin formation: Pass
-------------------------------------------------------------
```